

Math 141, S10

Quiz 2

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(4 points) Find a formula for the inverse of the function

$$f(x) = 2x^3 + 4.$$

Solution: $y = 2x^3 + 4 \Rightarrow y - 4 = 2x^3$

$$x^3 = \frac{y-4}{2}$$

$$x = \sqrt[3]{\frac{y-4}{2}}$$

Thus, the inverse function is

$$y = \sqrt[3]{\frac{x-4}{2}} \quad \text{or} \quad f^{-1}(x) = \sqrt[3]{\frac{x-4}{2}}$$

Problem 2.(6 points) Determine the following limits:

(a) $\lim_{x \rightarrow 1} \sqrt{x^2 + x + 1}$

$$= \sqrt{\lim_{x \rightarrow 1} (x^2 + x + 1)} = \sqrt{1^2 + 1 + 1} = \sqrt{3}$$

(b) $\lim_{x \rightarrow 3^-} \frac{x+1}{x-3}$

$$= -\infty$$